

## 1. Project Overview

Project FORESIGHT is an AI-powered demand forecasting and inventory intelligence platform designed to help businesses optimize inventory management and improve supply chain decisions. The platform uses historical sales data, machine learning algorithms, and business intelligence techniques to forecast product demand, identify stock risks, and generate inventory recommendations through an interactive Streamlit dashboard.

The system enables organizations to reduce inventory costs, prevent stock shortages, avoid overstock situations, and make data-driven inventory planning decisions.

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## 2. Problem Statement

Many businesses still manage inventory manually or rely on historical assumptions instead of predictive analytics. This often leads to:

- Stockouts
- Overstocking
- Increased inventory cost
- Lost sales
- Poor demand planning

Project FORESIGHT addresses these challenges by combining machine learning with inventory intelligence.

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## 3. Objectives

- Forecast future product demand using Machine Learning.
  - Analyze historical sales data.
  - Detect stockout and overstock risks.
  - Generate reorder recommendations.
  - Build an interactive business dashboard.
  - Support data-driven inventory planning.
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## 4. Technology Stack

### Programming Language

- Python

### Libraries

- Pandas
- NumPy
- Matplotlib
- Plotly
- Scikit-learn
- Joblib
- Streamlit

### Machine Learning

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

### Development Tools

- Visual Studio Code
- Git
- GitHub
- Streamlit Community Cloud

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## 5. Project Workflow

Raw Sales Dataset

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Data Cleaning

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Exploratory Data Analysis (EDA)

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Feature Engineering

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Machine Learning Model

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Demand Forecasting

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Inventory Intelligence

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Interactive Dashboard

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Deployment

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## **6. Dataset**

Dataset Used:

Chocolate Sales Dataset

Main Attributes:

- Sales Person
- Country
- Product
- Date
- Amount
- Boxes Shipped

Generated Dataset:

Inventory Dataset

Additional Attributes:

- Current Stock
  - Lead Time
  - Safety Stock
  - Reorder Point
  - Inventory Status
  - Recommended Order Quantity
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## **7. Project Phases**

### **Phase 1 – Data Cleaning**

Tasks Performed:

- Loaded dataset
- Removed duplicate records
- Checked missing values
- Converted Date to datetime
- Converted Amount to numeric
- Saved cleaned dataset

Output:

cleaned\_sales.csv

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### **Phase 2 – Exploratory Data Analysis**

Performed:

- Total Revenue Analysis
- Product-wise Revenue
- Country-wise Revenue
- Monthly Sales Trend

- Salesperson Performance
- Product Demand Analysis

Charts Created:

- Bar Charts
  - Line Charts
  - Pie Charts
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### **Phase 3 – Feature Engineering**

Created Features:

- Year
- Month
- Day
- Day of Week
- Quarter
- Weekend
- Product Encoding
- Country Encoding
- Salesperson Encoding
- Lag Features
- Rolling Average Features
- Revenue per Box

Output:

featured\_sales.csv

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### **Phase 4 – Machine Learning**

Algorithms Used:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

Evaluation Metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- $R^2$  Score

Best Model:

Random Forest Regressor

Saved Model:

demand\_model.pkl

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## **Phase 5 – Inventory Intelligence**

Implemented:

- Average Daily Demand
- Reorder Point Calculation
- Inventory Status Classification
- Stockout Detection
- Overstock Detection
- Recommended Order Quantity

Inventory Status:

- Healthy
- Reorder
- Overstock

Generated Report:

inventory\_report.csv

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## **Phase 6 – Interactive Dashboard**

Developed using Streamlit.

Features:

- KPI Cards
  - Revenue Trend
  - Monthly Sales Analysis
  - Product-wise Revenue
  - Country-wise Revenue
  - Inventory Status Table
  - Reorder Alerts
  - Overstock Alerts
  - AI Demand Prediction
  - Product Filter
  - Download Inventory Report
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## **8. Results**

The system successfully:

- Forecasts product demand.
  - Detects inventory risks.
  - Suggests reorder quantities.
  - Displays business insights using an interactive dashboard.
  - Supports better inventory planning.
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## **9. Applications**

- Retail Stores
  - Supermarkets
  - Warehouses
  - Supply Chain Companies
  - E-commerce Platforms
  - Manufacturing Industries
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## **10. Future Enhancements**

- Deep Learning Forecasting (LSTM)
  - Real-Time Database Integration
  - Weather-Based Demand Prediction
  - Email Alerts
  - Mobile Application
  - Multi-Warehouse Inventory Management
  - Barcode & QR Code Integration
  - ERP Integration
  - AI Chat Assistant
  - Cloud Database Support
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## **11. Conclusion**

Project FORESIGHT demonstrates how Artificial Intelligence and Machine Learning can transform traditional inventory management into a smart decision-support system. By integrating demand forecasting, inventory analysis, and interactive business intelligence, the platform enables organizations to improve operational efficiency, reduce costs, and make informed inventory decisions. This project also showcases practical skills in Python, data analytics, machine learning, visualization, and web application development, making it a strong portfolio project for data analytics and AI roles.

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